Slides for Patryk's Notes

Patryk Kozlowski

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Outline

1. Verification of LDA

2. Discussion of next steps

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Correlation energy per electron

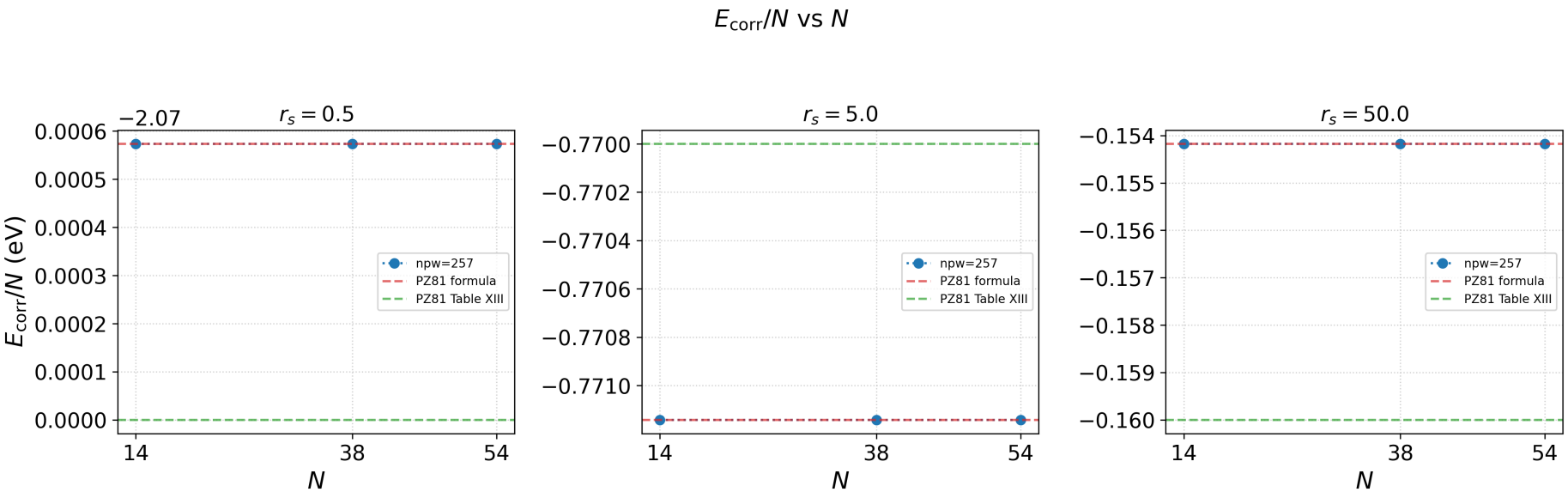
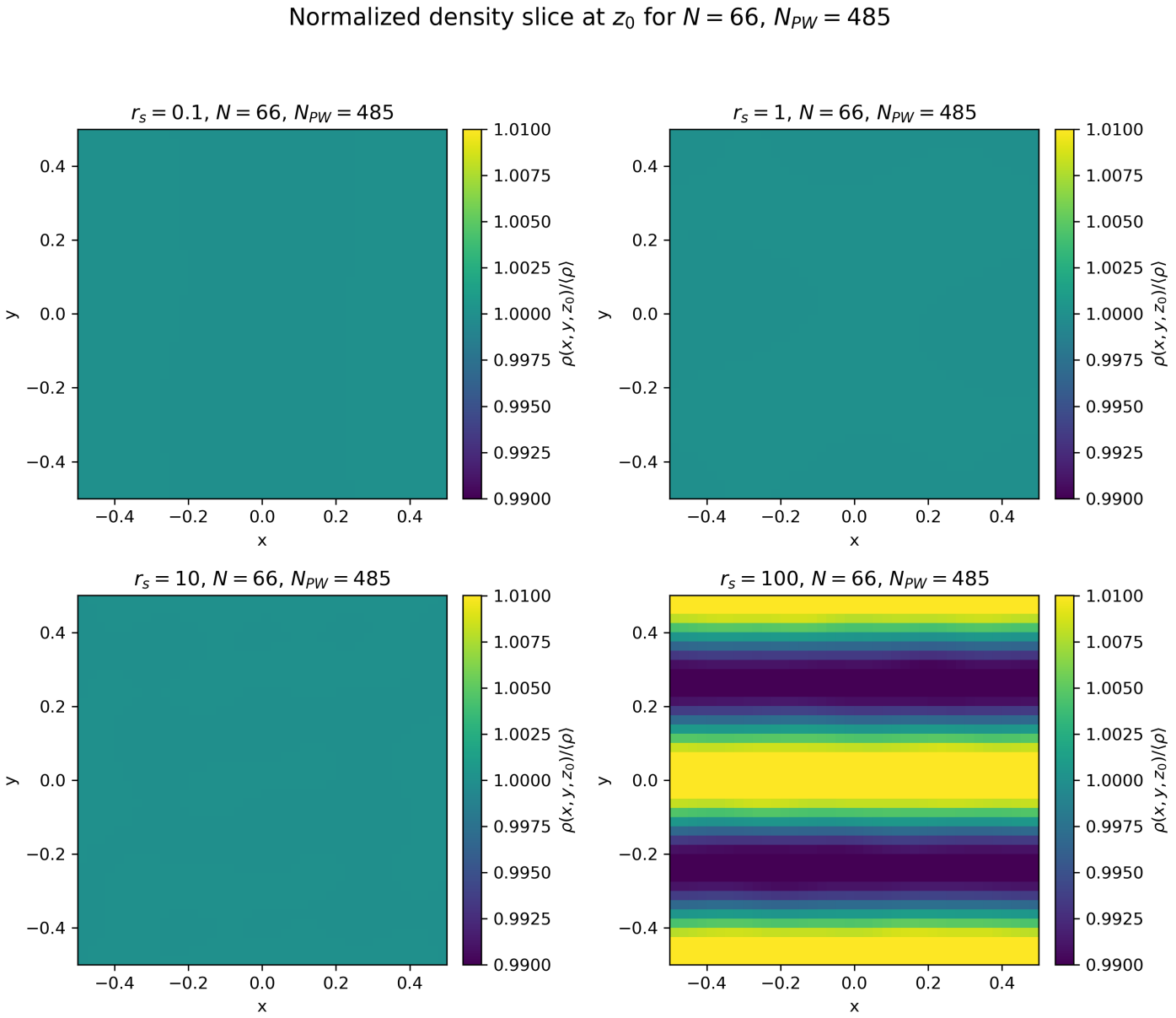
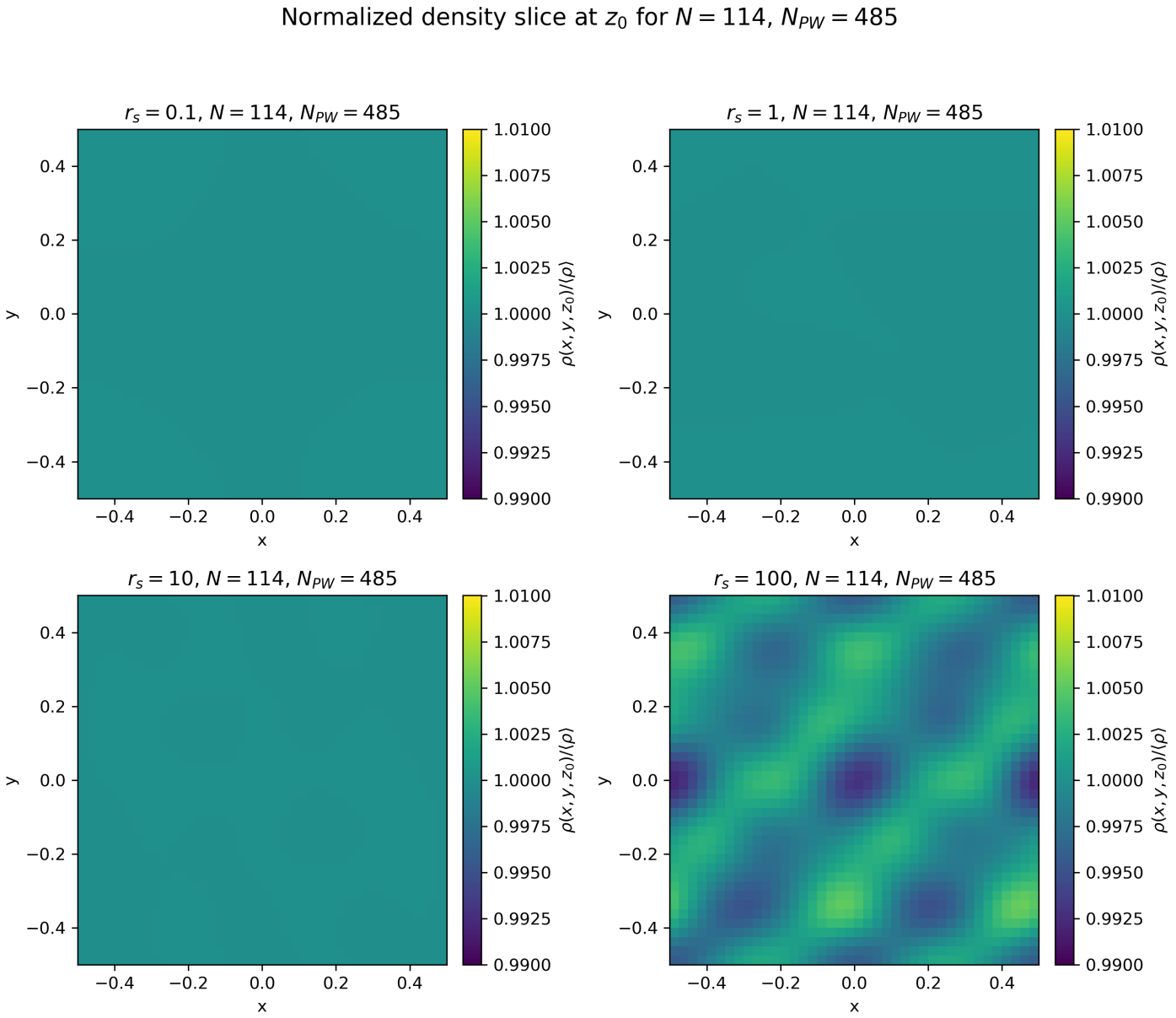


Figure: To be compared with PZ81, Table XIII.

Density as a function of r_s



Density as a function of r_s continued



Analysis of density slices

Inhomogeneity at low density is observed, as expected.

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Implementations of GW

- Implement GW with AC in Python; easiest to debug and check against PySCF that way.
- Nemo has been working on implementing GW with CD into C++.
- Once I verify that my Python implementation of GW with AC is correct, the QCPBC infrastructure will already be there to add GW with AC also.